

Eric Thomas Schneider

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Education

Virginia Tech

May 2029 (expected)

Computer Science (PhD)

Advisor: Dan Williams

GPA: 4/4

Select courses (all grad): Binary Exploitation, Research Methods, Advanced Linux Kernel Programming, Networking, Machine Learning, Agentic Systems.

Select courses (in progress): Agentic and Operating Systems.

University of North Carolina at Chapel Hill

May 2024

Computer Science (MS)

Select courses (all grad): Research Topics in Security, Computer Vision, Hardware Security and Side Channels.

University of North Carolina at Chapel Hill

May 2023

Computer Science (BS), Mathematics (BS)

GPA: 3.632/4

Select courses: Web, Databases, Compilers, Programming Languages, Algorithms, Operating Systems, Operating System Implementation (grad), Digital Logic, Research Topics in Operating Systems (grad), Software Engineering Laboratory, Differential Equations, Linear Algebra, Probability, Real Analysis, Numerical Analysis.

Research

ROSA (VT Computer Science Department)

Aug 2024-present

- Research trusted execution environments, sandboxing, LLMs, and eBPF under Prof. Dan Williams.
- Lead paper on sandboxing timing of an untrusted LLM application.
- Contributed to research on eBPF termination.
- Explore CPU/GPU covert channels and sandboxing in the context of confidential computing and LLMs, using AMD SEV-SNP, Intel TDX, NVIDIA CC.
- Explored media attestation using ARM CCA.
- Served on ASPLOS 2026 Artifact Evaluation Committee.
- Served on ASPLOS 2025 Artifact Evaluation Committee.

OSCAR (UNC Computer Science Department)

Jan 2023-May 2024

- Researched optimization of Gramine library OS using Intel SGX2 under Prof. Donald Porter.
- Contributed to paper, *Adaptive and Efficient Dynamic Memory Management for Hardware Enclaves* (published at SYSTOR '25).
- Managed Gramine benchmarking.
- Wrote literature review on mechanisms of memory protection.
- Explored port of Keystone TEE to uClinux.
- Served on ASPLOS 2024 Artifact Evaluation Committee.

ADAER (UNC Computer Science Department)

May 2023-Dec 2023

- Researched asynchronous video representation under Andrew Freeman, Prof. Ketan Mayer-Patel, and Prof. Montek Singh, funded by the Laboratory for Analytical Sciences (LAS) in summer 2023.
- Implemented transcoder bandwidth limiting feature in Rust.
- Implemented logging feature to Rust implementation of DVS Fast algorithm.
- Ported ADAER to Windows and improved CI using GitHub Actions.
- Wrote a survey paper.

Publications

Adaptive and Efficient Dynamic Memory Management for Hardware Enclaves

Vijay Dhanraj, Harpreet Singh Chawla, Daniel Manila, **Eric Thomas Schneider**, Erica Fu, Mona Vij, Chia-Che Tsai, Donald E. Porter

ACM International Systems and Storage Conference (SYSTOR), 2025.

Teaching

Teaching Assistant (VT Computer Science Department)

Aug 2025-present

- Tutor students in Computer Systems (CS 3214) under Prof. Back, Prof. Williams, and Prof. Jian (fall 2026).
- Tutor students in Data Structures and Software Design (CS 2114) under Prof. Margaret Ellis and Prof. Mohammed Farghally (fall 2025).
- Lead labs.
- Grade classwork, respond to student questions on Piazza.

Teaching Assistant (UNC Computer Science Department)

Aug 2023-May 2024

- Tutored students in Systems Fundamentals (COMP 211) under Prof. Brent Munsell and Operating Systems (COMP 530) under Prof. Donald Porter (fall 2023).
- Tutored students in Compilers (COMP 520) under Syed Ali (spring 2024).
- Managed quizzes, midterms, and the final.
- Managed and contributed towards labs and associated autograders, using Bash, Python, C, and Java.
- Graded classwork, responded to student questions on Piazza and CampusWire.

Undergraduate Teaching Assistant (UNC Computer Science Department)

Aug 2020-May 2023

- Tutored students in Systems Fundamentals (COMP 211) for 5 semesters under Prof. Kris Jordan (fall 2020), Prof. Ketan-Mayer Patel (spring 2022), and Prof. Brent Munsell (spring 2021, fall 2021, spring 2023).
- Tutored students in Operating Systems (COMP 530) for 1 semester (fall 2022) under Prof. Donald Porter.
- Managed quizzes, midterms, and the final.
- Managed, contributed towards, and wrote labs and associated autograders, using Bash, Python, and C.
- Graded classwork, responded to student questions on Piazza, CampusWire, and GroupMe.
- Collaborated to create a website to archive course material, using HTML, CSS and Bulma.

Industry Experience

Lucid Computing

Research Scientist Intern

May 2026-August 2026

- Researched AI safety, GPU fingerprinting, and trusted execution environments.
- Contributed towards grant proposals.
- Prototyped multi-cloud spot VM orchestrator for Lucid platform.

Nutanix

Software Developer Intern for Nutanix Database Services (NDB)

May 2024-August 2024

- Contributed to support for Nutanix Object Storage and Prism Central in NDB using Spring and React.js.
- Implemented Object Store storage consumption monitoring feature in NDB.

VMware

Software Developer Intern for VMware Telco Cloud Platform RAN

May 2022-August 2022

- Separated complex Flink-based streamer microservice into session collator and KPI composer microservice to increase product reliability, decrease complexity, and support VMware Centralized RIC, a non-real time RAN Intelligence Controller.

Software Developer Intern for VMware Uhana

May 2021-August 2021

- Migrated real-time streaming decoder microservice to Flink from Kafka Streams in Java, used to decode raw data from Kafka into protobufs, processing >1 million messages/second in production.

Skills and Interests

- Software engineering, AI, hardware, teaching, research.
- Applications/scripting: Java, Python, Ruby, Groovy, SQL, Lisp, Lua, C#, MATLAB, Mathematica.
- Systems: C, Assembly (x86, MIPS, RISC-V), Rust, C++, CUDA, Verilog.
- Web: MediaWiki, JavaScript/TypeScript, HTML, CSS, Node, Vue, Nuxt, React, WebGL.
- Tools: Bash, Git, Docker, Kubernetes, Make, Kafka, Kafka Streams, LaTeX.
- Mathematics (applied, calculus, algebra, proofs).
- Running (7 marathons), triathlon, weight training, international affairs, Spanish (basic).

Notable Projects

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| • Official FTB Wiki | Administer/edit wiki (>55k edits), using wikitext, Lua, JavaScript, CSS |
| • NeuroRuler | Contributed to Python tool to measure head circumference of MRIs |
| • JOS | Implemented portions of operating system in C/x86 Assembly |
| • MagnumVM | Created process virtual machine in Rust, winning 1st in small hackathon |
| • Psil | Created medium-sized Lisp-like programming language in Rust |
| • miniJava | Created compiler for Java subset in Java |
| • Tar Heel Calendar | Created backend for calendar website using express.js and MariaDB |
| • Personal website | Created website, originally in plain HTML/CSS/JS, later Vue/TypeScript |
| • Long Fall Boots | Ported/maintained popular Minecraft mod (~18m downloads) in Java |
| • ATT-9001 | Created tool for scraping tile translations for wiki in Ruby |
| • ESAEBSD | Created IRC/Discord bots used for automated wiki tasks in Ruby |
| • Flaxbeard's Steam Power | Contributed to popular Minecraft mod (~1m downloads) in Java |
| • Nuclear Control 2 | Ported/maintained popular Minecraft mod (~6m downloads) in Java |